The Box (Area) Method

Multiplying and Factoring Binomials

Class Notes · Organized Lesson with Worked Examples

Name: _____ **Date:** _____

Key Idea

The big idea. The area of a rectangle is (length) × (width). If each side is a *binomial*, split the rectangle into four smaller rectangles — one for each pair of terms. The four little areas add up to the answer.

We will learn the box in **two directions**:

- **Forward** (Part 1): given the two sides, *multiply* to find the area.
- **Backward** (Parts 2–3): given the area, *factor* to find the two sides.

Learning the forward direction first is what makes the backward direction make sense.

Part 1 • Forward — Multiplying Two Binomials

1.1 Simple case: leading coefficient $= 1$

☞ Worked Example

Multiply $(x+4)(x+5)$.

	x	$+4$
x	x^2	$4x$
$+5$	$5x$	20

x and $+4$ across the top	first factor
x and $+5$ down the side	second factor
cell = (row) × (column)	area of each piece
$x^2 + 4x + 5x + 20$	add all four cells
$= x^2 + 9x + 20$	combine middle terms

Notice: the two *diagonal* cells, $4x$ and $5x$, are the ones that combine into the middle term $9x$. Remember this — it is the key to factoring later.

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1.2 General case: leading coefficient $\neq 1$

☞ Worked Example

Multiply $(2x+3)(5x+7)$.

	$2x$	$+3$
$5x$	$10x^2$	$15x$
$+7$	$14x$	21

$$\begin{aligned} &10x^2+15x+14x+21 && \text{add the four cells}\\ &=10x^2+29x+21 && \text{combine } 15x+14x \end{aligned}$$

The box is just the distributive property done twice:

$$(2x+3)(5x+7)=(2x+3)(5x)+(2x+3)(7)=10x^2+15x+14x+21.$$

1.3 The general pattern

☞ Worked Example

Multiply $(ax+b)(cx+d)$ in general.

	ax	$+b$
cx	acx^2	bcx
$+d$	adx	bd

$$(ax+b)(cx+d)=acx^2+(ad+bc)x+bd$$

Key Idea

The corner property. In every box, multiply the two *corner* cells and compare with the two *diagonal* cells:

$$\underbrace{(acx^2)(bd)}_{\text{corners}}=abcd\,x^2 \qquad \text{and} \qquad \underbrace{(bcx)(adx)}_{\text{diagonals}}=abcd\,x^2.$$

They are **always equal**. And the two diagonal cells *add* to the middle term.

So to factor ax^2+bx+c, we need two terms whose **product** is $(ax^2)(c)=ac\,x^2$ and whose **sum** is bx. That is exactly what the "X" diagram organizes.

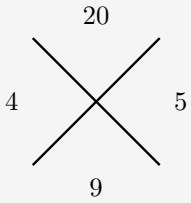
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Part 2 • Backward — Factoring when the Leading Coefficient is 1

☞ Worked Example

Factor $x^2 + 9x + 20$.

	x	$+4$
x	x^2	$4x$
$+5$	$5x$	20



top = product = 20
bottom = sum = 9
sides = the two numbers

Place x^2 and 20 in the corners	what we are given
need product 20, sum 9 $\Rightarrow$ 4 and 5	fill the diagonal: $4x$ and $5x$
read the sides: $x{+}4$ and $x{+}5$	take the GCF of each row/column

$$\boxed{x^2 + 9x + 20 = (x + 4)(x + 5)}$$

Key Idea

Shortcut when $a = 1$. You only need two numbers with **product** $= c$ and **sum** $= b$. Then $x^2 + bx + c = (x + p)(x + q)$ directly — no grouping required.

Part 3 • Backward — Factoring when the Leading Coefficient is not 1

3.1 Why guessing is slow

To factor $6x^2 + 13x + 5$ by trial and error you would test every pairing:

Guess	Expands to	Middle term
$(x + 1)(6x + 5)$	$6x^2 + 5x + 6x + 5$	$11x$ ✗
$(6x + 1)(x + 5)$	$6x^2 + 30x + x + 5$	$31x$ ✗
$(2x + 1)(3x + 5)$	$6x^2 + 10x + 3x + 5$	$13x$ ✓

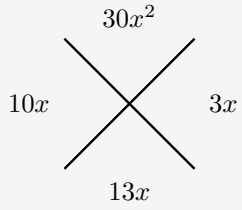
The X method finds the right one *directly*, with no guessing.

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3.2 The X method (product–sum), then the box

☞ Worked Example

Factor $6x^2 + 13x + 5$.



	$3x$	$+5$
$2x$	$6x^2$	$10x$
$+1$	$3x$	5

$ac = 6 \cdot 5 = 30$
need product 30, sum 13
$\Rightarrow$ 10 and 3

$$\begin{aligned} 6x^2 + 13x + 5 &= 6x^2 + 10x + 3x + 5 && \text{split the middle term using 10 and 3} \\ &= 2x(3x + 5) + 1(3x + 5) && \text{GCF of each row of the box} \\ &= (2x + 1)(3x + 5) && \text{factor out the common } (3x + 5) \end{aligned}$$

Check by re-multiplying: $(2x + 1)(3x + 5) = 6x^2 + 10x + 3x + 5 = 6x^2 + 13x + 5$ ✓

Key Idea

Reading the sides off the box.

- Take the **GCF of the top row** → first term of one factor.
- Take the **GCF of the bottom row** → second term of that factor.
- Take the **GCF of each column** → the other factor.

For the box above: rows give $2x$ and $+1$; columns give $3x$ and $+5$.

✗ Common Mistakes

- **Using c instead of ac.** When $a \neq 1$ the product you need is $a \cdot c$, not just c. For $6x^2 + 13x + 5$ it is 30, not 5.
- **Forgetting the signs.** The product's sign tells you whether the two numbers have the same sign $(+)$ or opposite signs $(-)$.
- **Putting the diagonal terms in the corners.** x^2 and the constant go in *opposite corners*; the two x-terms go in the other diagonal.
- **Stopping too early.** Always re-multiply to check.

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Part 4 • Summary — The Whole Method on One Page

Key Idea

To MULTIPLY $(ax+b)(cx+d)$**:**

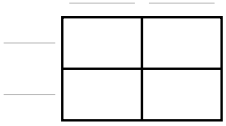
1. Write one factor across the top, the other down the side.
2. Fill each cell with (row) $\times$ (column).
3. Add all four cells and combine the two middle terms.

To FACTOR ax^2+bx+c**:**

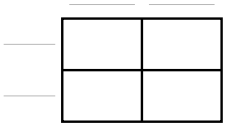
1. Factor out any **GCF** of all three terms first.
2. Compute $a \cdot c$. Find two numbers with product ac and sum b (use the X).
3. Put ax^2 and c in opposite corners; put the two x-terms in the other diagonal.
4. Take GCFs of the rows and columns to read off the two factors.
5. **Check** by multiplying back.

Quick Check — try these before the worksheet

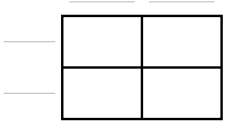
1. Multiply $(x+3)(x+6)$ using a box.



2. Factor $x^2+8x+15$.



3. Factor $2x^2+7x+3$. (Remember: $ac = 2 \cdot 3 = 6$.)



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Answer Key

[title=Quick Check — Answers] **1.** Cells $x^2,\ 6x,\ 3x,\ 18 \Rightarrow \mathbf{x^2 + 9x + 18}$. **2.** Product 15, sum $8 \Rightarrow 3, 5$; $(\mathbf{x + 3})(\mathbf{x + 5})$. **3.** $ac = 6$, sum $7 \Rightarrow 6, 1$; $2x^2 + 6x + x + 3 = 2x(x + 3) + 1(x + 3) = (\mathbf{2x + 1})(\mathbf{x + 3})$.

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Practice Worksheet

Multiplying and Factoring Binomials

Key Idea

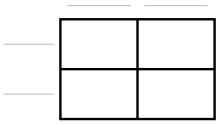
Reminders. Multiply: fill each cell with (row)×(column), then add. Factor: find two numbers with **product** $a \cdot c$ and **sum** b, split the middle term, then take GCFs of the rows and columns. **Always check by multiplying back.**

Difficulty: ●○○ Easy ●●○ Medium ●●● Hard

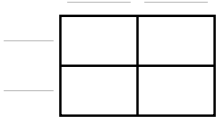
Practice A • Multiplying Binomials with a Box

Draw and fill the box, then write the product in standard form.

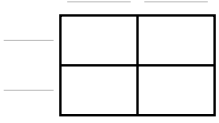
1. ●○○ **Easy** $(x+2)(x+6)$



2. ●○○ **Easy** $(x+3)(x-5)$

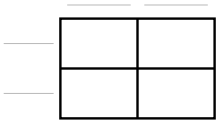


3. ●●○ **Medium** $(2x+1)(x+4)$

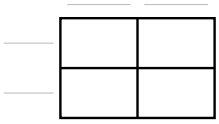


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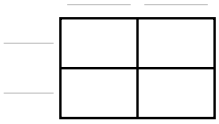
4. ●●○ **Medium** $(3x+2)(2x+5)$



5. ●●● **Hard** $(4x-3)(2x+7)$



6. ●●● **Hard** $(5x-2)(3x-4)$

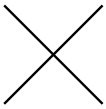


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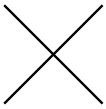
Practice B • Factoring when the Leading Coefficient is 1

Find two numbers with product c and sum b. Use the X, then the box.

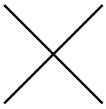
1. ●○○ **Easy** $x^2 + 7x + 12$



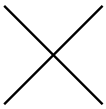
2. ●○○ **Easy** $x^2 + 10x + 21$



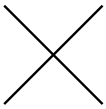
3. ●●○ **Medium** $x^2 - 9x + 20$



4. ●●○ **Medium** $x^2 + 2x - 15$

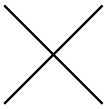


5. ●●○ **Medium** $x^2 - 3x - 28$



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6. ●●● **Hard** $x^2 - 16$ *(what is the middle term?)*

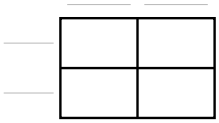


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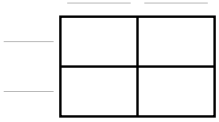
Practice C • Factoring when the Leading Coefficient is not 1

Compute $a \cdot c$ first. Product $= ac$, sum $= b$.

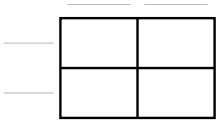
1. ●○○ **Easy** $2x^2 + 7x + 3$



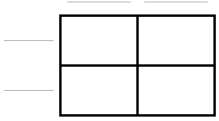
2. ●●○ **Medium** $3x^2 + 10x + 8$



3. ●●○ **Medium** $6x^2 + 17x + 5$

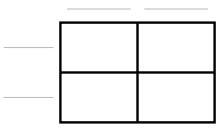


4. ●●○ **Medium** $4x^2 - 4x - 15$

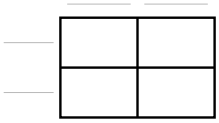


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5. ●●● **Hard** $10x^2 + 29x + 21$



6. ●●● **Hard** $6x^2 - 7x - 3$



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Practice D • Reasoning & Applications

1. ●●○ **Medium** A rectangle has area $2x^2 + 11x + 12$ square units. One side measures $2x + 3$. Find the other side.

2. ●●○ **Medium** The product $12x^2 + bx + 15$ factors as $(3x + 5)$ times another binomial. Find the other binomial and the value of b.

3. ●●● **Hard** Complete the box. Two cells are filled in; find the missing cells *and* both binomial factors.

$6x^2$	
	-10

Hint: there is more than one correct answer. Give one and check it.

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4. ●●● **Hard** Marcus factored $8x^2 + 10x - 3$ and got $(4x - 1)(2x + 3)$. Multiply his answer back out. Is he right? If not, find the correct factorization and explain his mistake.

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Complete Answer Key

with the product–sum reasoning for each problem

Answer Key

[title=Part 1 — Multiplying Binomials]

1. Cells $x^2,\ 6x,\ 2x,\ 12 \Rightarrow \mathbf{x^2 + 8x + 12}$
2. Cells $x^2,\ -5x,\ 3x,\ -15 \Rightarrow \mathbf{x^2 - 2x - 15}$
3. Cells $2x^2,\ 8x,\ x,\ 4 \Rightarrow \mathbf{2x^2 + 9x + 4}$
4. Cells $6x^2,\ 15x,\ 4x,\ 10 \Rightarrow \mathbf{6x^2 + 19x + 10}$
5. Cells $8x^2,\ -6x,\ 28x,\ -21 \Rightarrow \mathbf{8x^2 + 22x - 21}$
6. Cells $15x^2,\ -6x,\ -20x,\ 8 \Rightarrow \mathbf{15x^2 - 26x + 8}$

Answer Key

[title=Part 2 — Factoring, Leading Coefficient = 1]

1. Product 12, sum $7 \Rightarrow 3, 4$. $\quad \mathbf{(x+3)(x+4)}$
2. Product 21, sum $10 \Rightarrow 3, 7$. $\quad \mathbf{(x+3)(x+7)}$
3. Product 20, sum $-9 \Rightarrow -4, -5$ (both negative). $\quad \mathbf{(x-4)(x-5)}$
4. Product -15, sum $2 \Rightarrow 5, -3$ (opposite signs). $\quad \mathbf{(x+5)(x-3)}$
5. Product -28, sum $-3 \Rightarrow -7, 4$. $\quad \mathbf{(x-7)(x+4)}$
6. Write it as $x^2 + 0x - 16$: product -16, sum $0 \Rightarrow 4, -4$. $\quad \mathbf{(x-4)(x+4)}$ — a *difference of squares.*

Answer Key

[title=Part 3 — Factoring, Leading Coefficient $\neq 1$]

1. $ac = 6$, sum $7 \Rightarrow 6, 1$: $\ 2x^2 + 6x + x + 3 = 2x(x+3) + 1(x+3) = \mathbf{(2x+1)(x+3)}$
2. $ac = 24$, sum $10 \Rightarrow 6, 4$: $\ 3x^2 + 6x + 4x + 8 = 3x(x+2) + 4(x+2) = \mathbf{(3x+4)(x+2)}$
3. $ac = 30$, sum $17 \Rightarrow 15, 2$: $\ 6x^2 + 15x + 2x + 5 = 3x(2x+5) + 1(2x+5) = \mathbf{(3x+1)(2x+5)}$
4. $ac = -60$, sum $-4 \Rightarrow -10, 6$: $\ 4x^2 - 10x + 6x - 15 = 2x(2x-5) + 3(2x-5) = \mathbf{(2x-5)(2x+3)}$
5. $ac = 210$, sum $29 \Rightarrow 14, 15$: $\ 10x^2 + 14x + 15x + 21 = 2x(5x+7) + 3(5x+7) = \mathbf{(2x+3)(5x+7)}$
6. $ac = -18$, sum $-7 \Rightarrow -9, 2$: $\ 6x^2 - 9x + 2x - 3 = 3x(2x-3) + 1(2x-3) = \mathbf{(2x-3)(3x+1)}$

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Answer Key

[title=Part 4 — Reasoning & Applications]

1. Factor the area: $ac = 2 \cdot 12 = 24$, sum $11 \Rightarrow 8, 3$; $2x^2+8x+3x+12 = 2x(x+4)+3(x+4) = (2x+3)(x+4)$. Since one side is $2x+3$, the other side is $\mathbf{x+4}$. *Check:* $(2x+3)(x+4) = 2x^2+11x+12$ ✓

2. The x^2 cells must give $12x^2$, so the other leading term is $4x$; the constants must give 15, so the other constant is 3. The other binomial is $\mathbf{4x+3}$. Multiplying: $(3x+5)(4x+3) = 12x^2+9x+20x+15 = 12x^2+29x+15$, so $\mathbf{b = 29}$.

3. One valid answer: put $6x^2$ and -10 in opposite corners. Since $ac = 6 \cdot (-10) = -60$, choose any pair with product -60, for example 15 and -4 (sum 11). The box becomes

$$6x^2+15x-4x-10 = 3x(2x+5)-2(2x+5) = (\mathbf{3x-2})(\mathbf{2x+5}),$$

   which expands to $6x^2+11x-10$. *Other answers are possible* — e.g. -12 and 5 give $(3x+5)(2x-2)$, i.e. $6x^2-6x-10$; any pair with product -60 works, and each gives a different middle term.

4. Multiply back: $(4x-1)(2x+3) = 8x^2+12x-2x-3 = 8x^2+10x-3$. **Marcus is correct!** ✓ His factorization checks out, which is exactly why re-multiplying is worth doing — it confirms a right answer just as clearly as it catches a wrong one. *(Verify with the X: $ac = 8 \cdot (-3) = -24$, sum $10 \Rightarrow 12$ and -2 $\Rightarrow 8x^2+12x-2x-3 = 4x(2x+3)-1(2x+3) = (4x-1)(2x+3)$.)*

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